Assignment 2: pg. 2 Kelly Nobles 2/24/26

→ Question 2: part B.

$\mu = 25.41$ $\sigma = 2.74$

$2(2.74) = 5.48$

$25.41 - 5.48 = 19.93$ $25.41 + 5.48 = 30.89$

$P(19.93 \le S \le 30.89)$

* All stock prices fall ✓Q2b the 2 std. dev. range.

therefore $\boxed{P(\text{w/in 2 std. dev.}) = 1.00}$

or

100% probability that stock price will be w/in 2 std. dev. of its mean 1yr from now

→ Question 3: Distribution of the option value

Price	Prob. f(x)	value	value	f(v)
20	0.05	0	0	✓Q3 0.47
21	0.06	0	$1	0.15
22	0.07	0	$2	0.13
23	0.08	0	$3	0.10
24	0.09	0	$4	0.10
25	0.12	0	$5	0.05
26	0.15	$1		1.0
27	0.13	$2		
28	0.10	$3		
29	0.10	$4		
30	0.05	$5		

= 0.47

sum

→ Question 4: Prob. will have positive value

Add $1 to $5 in ✓Q4 right-hand table from question 3

$\boxed{P(\text{pos. value}) = 0.53}$

or

53% probability stock will have pos. value